

Installation Instructions

NOTE: Read the entire instruction manual before starting the installation.

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SAFETY CONSIDERATIONS

Improper installation, adjustment, alteration, service, maintenance, or use can cause explosion, fire, electrical shock or other conditions which may cause personal injury or property damage. Consult a qualified installer, service agency, or your distributor or branch for information or assistance. The qualified installer or agency must use factory-authorized kits or accessories when modifying this product. Refer to the individual instructions packaged with the kits or accessories when installing.

Follow all safety codes. Wear safety glasses and work gloves. Use quenching cloths for brazing operations. Have fire extinguisher available. Read these instructions thoroughly and follow all warnings or cautions attached to the unit. Consult local building codes and National Electrical Codes (NEC) for special requirements.

It is important to recognize safety information. This is the safety-alert symbol . When you see this symbol on the unit and in instructions or manuals, be alert to the potential for personal injury.

Understand the signal words DANGER, WARNING, CAUTION, and NOTE. These words are used with the safety-alert symbol. DANGER identifies the most serious hazards which **will** result in severe personal injury or death. WARNING signifies hazards which **could** result in personal injury or death. CAUTION is used to identify unsafe practices, which **may** result in minor personal injury or

product and property damage. NOTE is used to highlight suggestions which **will** result in enhanced installation, reliability, or operation.

	WARNING: ELECTRICAL SHOCK HAZARD Failure to follow this warning could result in personal injury or death. Before installing or servicing system, always turn off main power to system. There may be more than one disconnect switch. Turn off accessory heater power if applicable.
	CAUTION: PERSONAL INJURY HAZARD Failure to follow this caution may result in personal injury. This coil contains Nitrogen precharge of up to 15 PSI. Release of this pressure through the center of the rubber plugs is required before removing the plugs.

IMPORTANT: Nitrogen can leak out through the hole that the needle pierced in the plugs. This does not indicate a leaking coil nor warrant return of the coil.

INTRODUCTION

Use this instruction manual to install CAPMP and CARMP indoor coils on multipoise furnaces. (See Fig. 1.) Models CAPMP/CARMP are enclosed in a casing.

NOTE: Models with tin-plated copper coils, "T" in the 11th position of the model number, are installed the same as standard copper coils.

INSTALLATION

PROCEDURE 1 — INSPECT EQUIPMENT

File claim with shipper if equipment is damaged or incomplete.

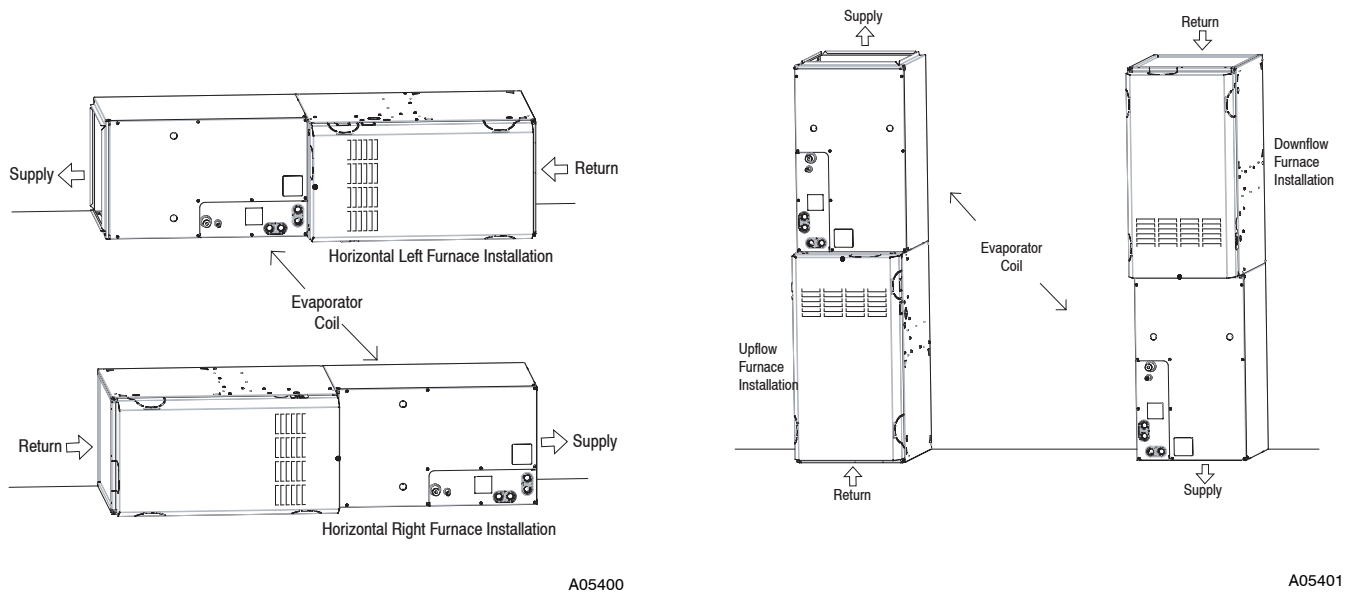
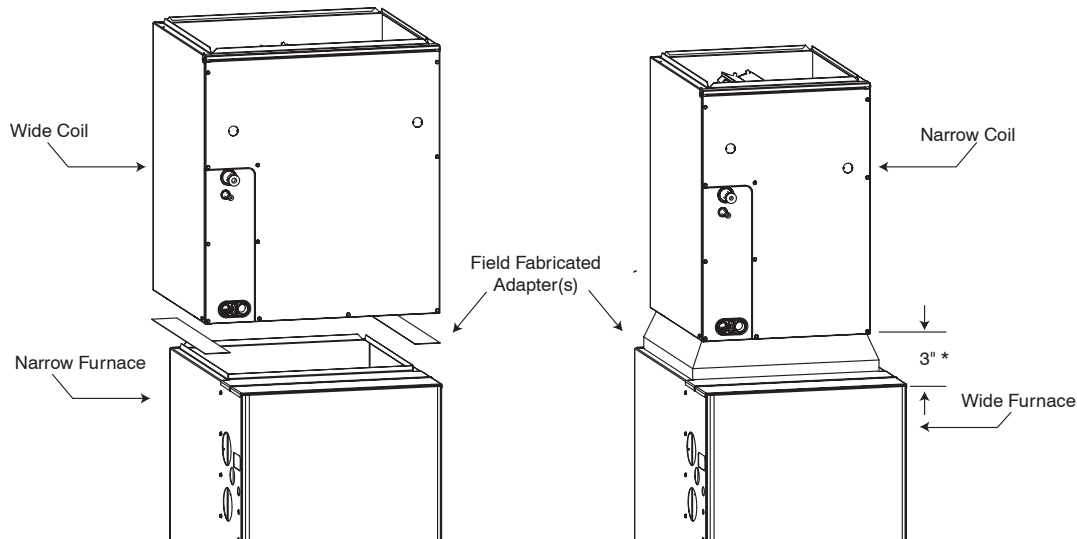


Fig. 1 - Typical Installation Multipoise Cased Coils



* CAUTION: DO NOT MOUNT COIL LOWER THAN 3" MINIMUM OR FURNACE FAILURE MAY RESULT

Fig. 2 - Adapter(s) Installation When Coil Overhangs or Underhangs Furnace

PROCEDURE 2 — SELECT INSTALLATION

For upflow application follow:

A. Upflow CAPMP/CARMP Coil Installation.

See Table 1 for dimensions. Note instructions for placement of coil casing on furnace.

For downflow application follow:

B. Downflow CAPMP/CARMP Coil Installation.

For horizontal application follow:

C. Horizontal CAPMP/CARMP Coil Installation.

PROCEDURE 3 — INSTALLATION OF EVAPORATOR COILS

A. Upflow CAPMP/CARMP Cased Coil Installation

The cased coil is designed to fit furnaces of the same width.

1. Set coil in place on upflow furnace discharge air opening.
2. Ensure coil is level for proper condensate drainage. Do not tip coil toward condensate drain. Coil casing need not be fastened or screwed to furnace.
3. When installing wider coil on narrow furnace, create field fabricated adapter. (See Fig. 2.)
4. When installing narrow coil on wide furnace, create field fabricated adapter. (See Fig. 2.)

NOTE: On upflow installations where the indoor coil is placed in an unconditioned space, a 6" wide piece of insulation should be applied and wrapped around the outside of coil casing and supply duct contact point.

B. Downflow CAPMP/CARMP Cased Coil Installation

IMPORTANT: Installing “A” coils rotated 90 degrees from the front of the furnace in downflow applications can cause water blow off or coil freeze up. This is due to the concentration of air on one coil slab or lack of air on the opposite coil slab. If the airflow is high due to ductwork or other causes, and there is a chance for water blow off, it is recommended that a 3-in. field-supplied adapter be placed between the coil and the furnace to allow the air to distribute evenly to both coil slabs. (See Fig. 3.)

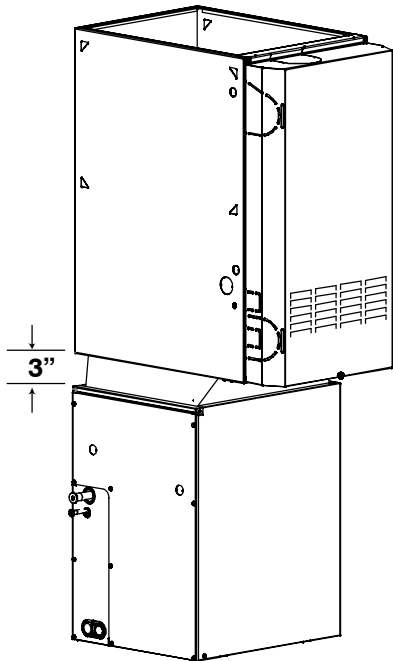


Fig. 3 - Downflow Installation With Coil Rotated 90°

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1. Set cased coil on supply duct opening.
2. Place field fabricated 3 in. adapter on coil casing. Adapter should be tapered to fit coil/furnace combination when one of them is larger than the other.
3. Set furnace on adapter.

NOTE: In downflow installation with a 4-way multipoise furnace, break off perforated duct flanges on furnace. See furnace installation instructions.

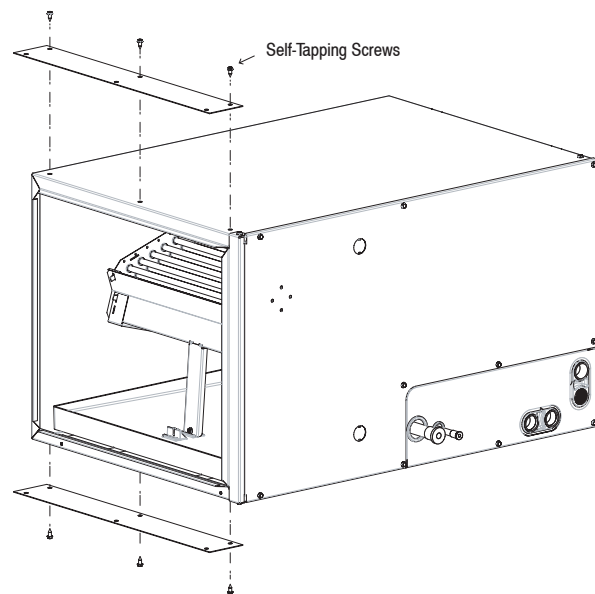
C. Horizontal CAPMP/CARMP Coil Installation

The unit can be installed on a work platform, secured to roof truss in attic, suspended from hangers on floor joists in crawl space, or installed on blocks. It is designed to allow airflow in either direction, to mate with horizontal-left or horizontal-right furnace installations.

Coil attachment to furnace.

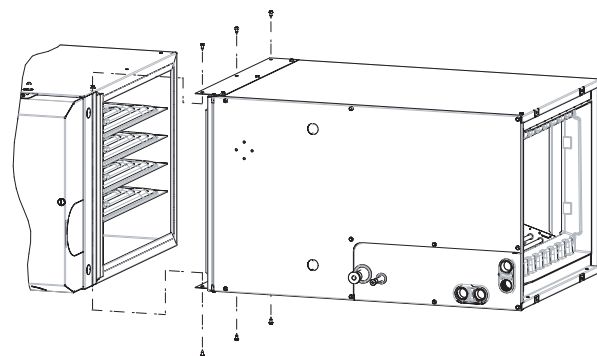
Horizontal Right Installation

1. Use field fabricated attachment plates to secure coil to furnace. (See Fig. 4.)
2. Use self-tapping screws to mount attachment plates to coil casing.
3. Connect furnace snugly against coil casing.
4. Use self-tapping screws to attach furnace. (See Fig. 5.)
5. Seal joint between coil casing and furnace to create air tight seal using locally approved materials.



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Fig. 4 - Horizontal Right Installation

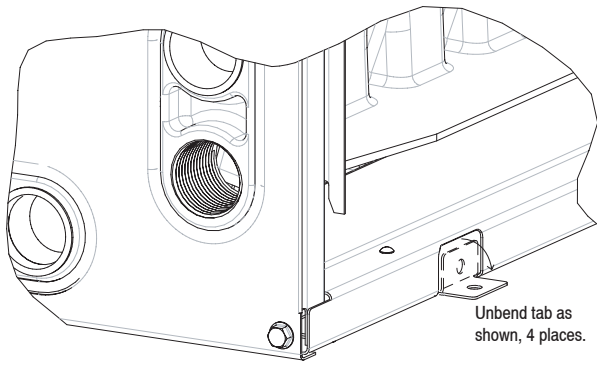


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Fig. 5 - Horizontal Right Installation

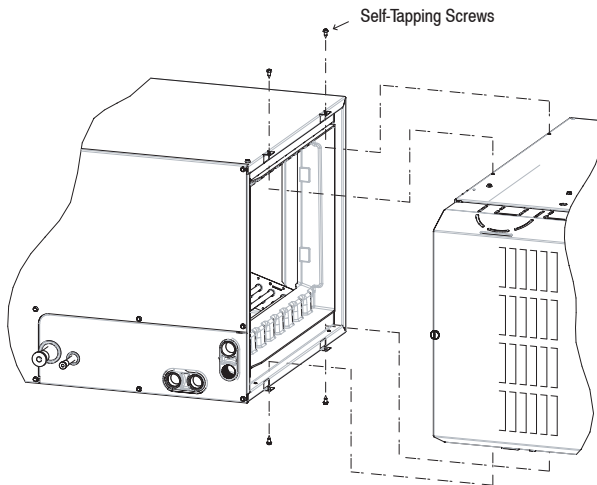
Horizontal Left Installation

1. Unbend the 4 tabs at the right side of the casing. (See Fig. 6.)
2. Connect furnace snugly against coil casing.
3. Use self-tapping screws to attach furnace. (See Fig. 7.)
4. Seal joint between coil casing and furnace to create air tight seal using locally approved materials.



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Fig. 6 - Horizontal Left Installation



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Fig. 7 - Horizontal Left Installation

PROCEDURE 4 — CONNECT REFRIGERANT PIPING

Use accessory tubing package or field-supplied tubing of refrigerant grade, see Product Data information for ordering. Suction tube must be insulated. Do not use damaged, dirty, or contaminated tubing because it may plug refrigerant flow-control device. ALWAYS evacuate the coil and field-supplied tubing before opening outdoor unit service valves.

PROCEDURE 5 — CONNECT REFRIGERANT, LIQUID, AND SUCTION LINES

For matched and mismatched systems, use line sizes recommended in outdoor unit Installation Instructions.

The coil can be connected to outdoor units using accessory tubing packages or field-supplied tubing of refrigerant grade. Always evacuate tubing and reclaim refrigerant when making connections or flaring tubing. Leak check connections before insulating entire suction line.

See Table 1 for coil connection tube size.

1. Remove cabinet door. Remove tubing plate with rubber grommets and slide plate with grommets onto the refrigerant lines (field line-set), away from braze joints.
2. Remove rubber plugs from coil stubs using a pulling and twisting motion. Hold coil stubs steady to avoid bending or distorting.
3. Wrap TXV and nearby tubing with a heat-sinking material such as a wet cloth.
4. Fit refrigerant lines into coil stubs. Wrap a heat sinking material such as a wet cloth behind braze joints.
5. Braze using a Sil-Fos or Phos-copper alloy.
6. After brazing, allow joints to cool. Slide tubing plate with rubber grommets over joints. Position tubing at center of each grommet to ensure an air seal around the tube.

CAUTION:

UNIT DAMAGE HAZARD

Failure to follow this caution may result in product damage.

To avoid valve damage to the refrigerant control device while brazing, valves must be wrapped with a heat-sinking material such as a wet cloth.

REFRIGERANT METERING DEVICE

CAPM Models:

These Coils have a factory installed hard shut-off TXV designed only for use with R-410A refrigerant. Use only with outdoor units designed for R-410A.

CARM Models:

These Coils have a factory installed hard shut-off TXV designed only for use with R-22 refrigerant. Use only with outdoor units designed for R-22.

NOTE: ALL TXV'S HAVE PRESET SUPERHEAT SETTINGS AND ARE FIELD **NON-ADJUSTABLE**.

Table 1—CAPMP/CARMP CASED COIL INFORMATION

MODEL NUMBER	TONNAGE	FLUSH FIT TO FURNACE WIDTH (IN.)	COIL CONNECTION TUBE SIZE (IN.)	
			Liquid	Suction
CA(P,R)MP1814A(C,T)A	1-1/2	14-3/16	3/8	5/8
CA(P,R)MP2414A(C,T)A	2	14-3/16	3/8	5/8
CA(P,R)MP2417A(C,T)A	2	17-1/2	3/8	5/8
CA(P,R)MP3014A(C,T)A	2-1/2	14-3/16	3/8	3/4
CA(P,R)MP3017A(C,T)A	2-1/2	17-1/2	3/8	3/4
CA(P,R)MP3614A(C,T)A	3	14-3/16	3/8	3/4
CA(P,R)MP3617A(C,T)A	3	17-1/2	3/8	3/4
CA(P,R)MP3621A(C,T)A	3	21	3/8	3/4
CA(P,R)MP4221A(C,T)A	3-1/2	21	3/8	7/8
CA(P,R)MP4224A(C,T)A	3-1/2	24-1/2	3/8	7/8
CA(P,R)MP4817A(C,T)A	4	17-1/2	3/8	7/8
CA(P,R)MP4821A(C,T)A	4	21	3/8	7/8
CA(P,R)MP4824A(C,T)A	4	24-1/2	3/8	7/8
CA(P,R)MP6021A(C,T)A	5	21	3/8	7/8
CA(P,R)MP6024A(C,T)A	5	24-1/2	3/8	7/8

Note: For the 3rd digit position in the model number; R = R-22 and P = Puron (R-410A)

Note: For the 11th digit position in the model number; C=Copper and T= Tin plated Copper (CAPMP Coils Only)

CAUTION:

UNIT DAMAGE HAZARD

Failure to follow this caution may result in product damage.



DO NOT BURY MORE THAN 36 IN. OF REFRIG-ERANT TUBING IN GROUND. If any section of tubing is buried, there must be a 6 in. vertical rise to the valve connections on the outdoor unit. If more than the recommended length is buried, re- frigerant may migrate to cooler buried section dur- ing extended periods of unit shutdown, causing refrigerant slugging and possible compressor damage at start-up.

PROCEDURE 6 — MAKE CONDENSATE DRAIN LINE CONNECTION

The coil is designed to dispose of accumulated water through built-in condensate drain fitting. Two 3/4 in. female threaded pipe con- nections are provide in each coil.

Install a trap in condensate line of coil, as close to the coil as possible. Make trap at least 3 in. deep and not higher than bottom of unit con- densate drain opening. (See Fig. 8.) Pitch condensate line to open drain or sump.

NOTE: The unit is shipped from the factory with the horizontal drain connections capped as for vertical installations. For horizontal installations, remove the caps and place them on the vertical drain connections to keep an airtight installation.

NOTE: If unit is located in or above a living space where damage may result from condensate overflow, a field-supplied, external condensate pan should be installed under the entire unit, and a secondary condensate line (with appropriate trap) should be run from the unit into the pan. Any condensate in this external

condensate pan should be drained to a noticeable place. As an alternative to using an external condensate pan, some localities may allow the running of a separate 3/4 in. condensate line (with appropriate trap) per local code to a place where the condensate will be noticeable. The owner of the structure must be informed that when condensate flows from secondary drain or external condensate pan, the unit requires servicing or water damage will occur.

NOTE: To avoid drainage problems, test the primary drain line by slowly pouring water into the pan. Check piping for leaks and proper condensate drainage. Using the secondary drain as explained in the previous note provides future protection against overflow due to a clogged primary drain.

NOTE: In applications where return air humidity levels stay at 70% or above for a prolonged period of time, condensation can form on the bottom of pan and drip.

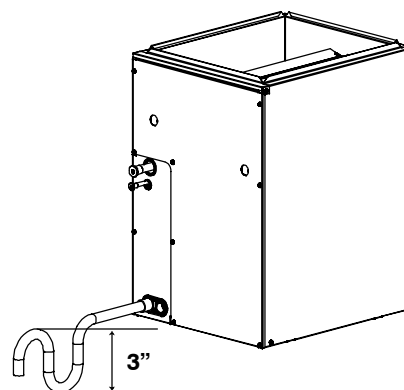


Fig. 8 - Condensate Trap

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